



RF-B-ARx BLE Beacon User Guide

Version 0.1.6

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1 Device Overview

There are two modes of Beacon broadcasting: connectable mode and unconnectable mode.

Connectable mode: The user can use the APP to connect the Beacon (enter a specific key when the user authentication function is enabled). After the connection, a specific command is issued to modify the beacon parameters on the corresponding channel.

Un-connectable mode: The Beacon broadcasts according to the data set by the user. The default is the connectable mode, and it can be configured for mode switching through serial AT commands.

Bluetooth Low Energy Beacon can be applied to indoor positioning, activity detection, asset tracking, etc. It can realize remote data management, and It can also configure the Beacon with different parameters through the mobile APP and the serial port respectively, so as to meet different applications.

The Beacon supports a variety of broadcasts, and each broadcast can be individually set for the broadcast period and transmit power. At the same time, the Beacon also has an accelerometer and a temperature and humidity sensor, and the two sensors can set the sampling period and other parameters respectively.



2 Beacon Parameters

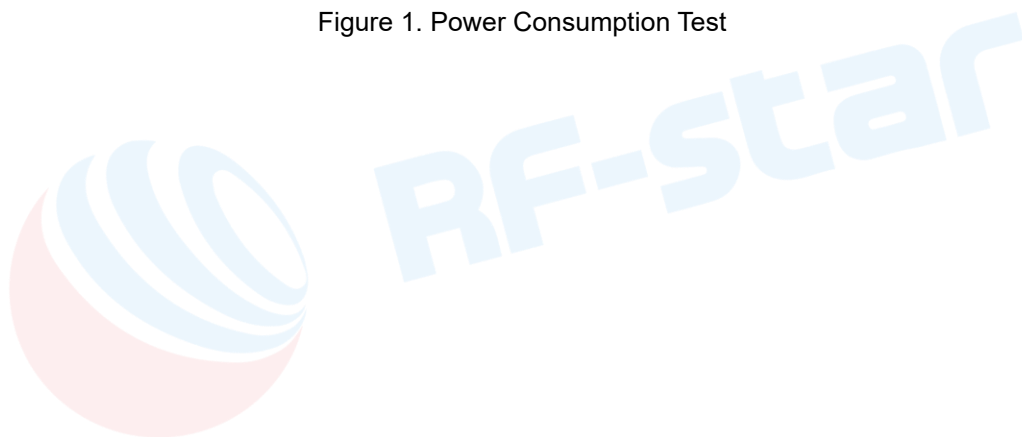
Table 1. Table of Parameters

Item	Value	Remarks
Model	RF-B-AR1	No sensors
	RF-B-AR2	Accelerometer sensor
	RF-B-AR3	Temperature and humidity sensor
	RF-B-AR4	Accelerometer & temperature and humidity sensors
Main IC	ATM2202	
BLE version	BLE5.0	Long range supported
Flash	1 MB	
RAM	128 KB	
ROM	256 KB	
Operating Voltage	1.7 V ~ 3.3 V	
TX Power	-20 dBm ~ +4 dBm	0 dBm is by default.
Broadcast Interval	1 s	
Transmission Distance	50 m	In an open air and sunny weather, @ 0 dBm
Average Power Consumption	7.6 μ A	1 s broadcast interval @ AR1
	13.1 μ A	1 s broadcast interval @ AR2
	14.98 μ A	1 s broadcast interval @ AR3
	21.88 μ A	1 s broadcast interval @ AR4
Battery	CR2477	10 years' service life under 1 s broadcast interval.
External Dimension	51.0 mm (ϕ) $\times$ 21.0 mm	Tolerance: $\pm$ 0.2 mm
Weight	29.6 g	AR1
	29.7 g	AR2
	29.8 g	AR3
	29.9 g	AR4
Waterproof Level	IP65	Splash-proof
Operating Temperature	-20 $^{\circ}$ C ~ +60 $^{\circ}$ C	Affected by the enclosure
Storage Temperature	-40 $^{\circ}$ C ~ +85 $^{\circ}$ C	Affected by the enclosure
Other Functions	OTA	
	Number of slots	Up to 10 different slots, including iBeacon, UID, URL, TLM.

Power consumption test is shown as below:



Figure 1. Power Consumption Test



3 LED and Button Specifications

Button	LED	Function	Remark
Press for 1 s under sleep mode	LED flash once	Boot up	
Press for 1 s under working mode	LED flash twice	Shut down	Boot-up mode is by default after power-on.
Press for 5 s under working mode	LED on for 2 s	Restore factory parameters	



4 Parameters Configuration

Table 2. Table of Parameter Configuration

Item	Default Settings
UUID (16 Bytes)	0x0112233445566778899AABBCCDDEEFF0
Major (2 Bytes)	1800
Minor (2 Bytes)	1286
Company ID (2 Bytes)	0x4C00
RSSI (2 Bytes)	-48 dBm
TX Power	0 dBm
Broadcast Interval	1 s
Broadcast Name	RFstar_XXXX (the suffix XXXX is the last two bytes of the MAC address)
Anti-hijacking Key (User Authentication)	Off by default
Connectable Mode	Connectable

5 BLE Protocol Specification (APP Interface)

Service UUID

6E400001B5A3F393E0A9E50E24DCCA9E

Enter Anti-hijacking Key UUID

Characteristics UUID	6E400002B5A3F393E0A9E50E24DCCA9E
Executable operations	Write
Remarks	When the user authentication function is enabled, an anti-hijacking key is required, and there is no return value. If the key is entered incorrectly or no anti-hijacking key is entered and the timeout period is reached, the connection is automatically disconnected. If the key is entered correctly, it will return OK\r\n. The timeout period can be configured by AT command, and the default value is 15 s.

AT Command Operation UUID

Characteristics UUID	6E400004B5A3F393E0A9E50E24DCCA9E
Executable operations	Write/Notify
Remarks	Input AT commands and then receive the command returns through this channel. Support all the commands in the AT command list, and any data without CRLF will be treated as commands.

6 Response Packet Data Format

6.1 Without Sensor

The broadcast type is iBeacon format by default.

The response packet data of the Beacon device (RF-B-AR1) without sensors is shown as follows. Service Data (Type: 0x16) includes the MAC address and can be acquired by iOS system:

Len	0x0C	Broadcast name length (1 byte)
Type	0x09	Broadcast type (1 byte)
Data	Name Data	Name (11 bytes)
Len	0x0C	Service data length (1 byte)
Type	0x16	Broadcast type (1 byte)
Data	0x1803	UUID (2 bytes)
	MAC	MAC address of the Beacon device (6 bytes)
	0x0708	Major (2 bytes)
	0x0506	Minor (2 bytes)
	0x00	TX power (1 byte)
	0x03E8	Broadcast Interval (2 bytes)
	0x51	Battery percentage (1 byte)

Raw data:

```
0x0201061AFF4C00021501122334455667
78899AABBCCDDEEFF007080506D00C0
95246737461725F4336344311160318FF3
ADAF6C64C070805060003E851
```

Details:

Beacon Data

LEN.	TYPE	VALUE
2	0x01	0x06
26	0xFF	0x4C0002150112233445566778899AABBCCDDEEFF007080506D0
12	0x09	0x5246737461725F43363443
17	0x16	0x0318FF3ADAF6C64C070805060003E851

LEN. - length of EIR packet (Type + Data) in bytes,
 TYPE - the data type as in <http://www.bluetooth.org/en-us/specification/assigned-numbers/generic-access-profile>

Broadcast Name

Service Data

The Service Data (Type:0x16) is 0318FF3ADAF6C64C070805060003E851, the data can be parsed as follows:

Service UUID: 0x1803

MAC address: FF:3A:DA:F6:C6:4C

Major: 0x0708, that is, 1800

Minor: 0x0506, that is, 1286

TX power: 0 dBm

Broadcast interval: 0x03E8, that is, 1000 ms

Battery percentage: 0x51, that is, 81%

6.2 With Sensors

The data sampled of the device with sensors is placed in the response packet data. Each broadcast has a response packet. The default data format of the response packet is as follows:

Len	0x0C	Broadcast name length (1 byte)
Type	0x09	Broadcast type (1 byte)
Data	Name Data	Name (11 bytes)
Len	0x10	Self-defined manufacturer data length (1 byte)
Type	0xFF	Manufacturer type (1 byte)
Data	0x5246	Manufacturer ID (2 bytes)
	Identification	Connectable identifier
	Identification	Boardcasting encryption identifier
	X-axis	Accelerometer sensor data (6 bytes), 2 bytes for each X-axis, Y-axis, Z-axis
	Y-axis	
	Z-axis	
	Temperature	Temperature and humidity sensor data (4 bytes), 2 bytes for each temperature and humidity
	Humidity	
	Battery capacity	Battery percentage (1 byte)

Raw data:

```
0x0201061AFF4C00021501122334455667
78899AABBCCDDEEFF007080506D00C0
95246737461725F3232313110FF52460100
00FA80EC40C03F09381B64
```



Details:

Beacon Data

LEN.	TYPE	VALUE
2	0x01	0x06
26	0xFF	0x4C0002150112233445566778899AABBCCDDEEFF007080506D0
12	0x09	0x5246737461725F32323131
16	0xFF	0x5246010000FA80EC40C03F09381B64

LEN. – length of EIR packet (Type + Data) in bytes,

TYPE – the data type as in <https://www.bluetooth.org/en-us/specification/assigned-numbers/generic-access-profile>

Broadcast Device Name

User-defined Data

OK

The sensor data field is 00FA80EC40C03F09381B, and the data is low byte first and the high byte after. According to the data format in the above table, it can be parsed as:

X-axis: FA00

Y-axis: EC80

Z-axis: C040 (unit: mg)

The highest bit indicates the sign, that is, when the accelerometer is a negative number, the highest bit is 1, you need to invert the data and add 1 to get the original data.

Temperature: 093F, which is 23.67 degrees Celsius. The highest bit indicates the sign, that is, when the temperature is negative, the highest bit is 1. You need to invert the data and add 1 to get the original data.

Humidity: 1B38, which is 69.68%

For example:

Raw data:

```
0x0201061AFF4C00021501122334455
66778899AABBCCDDEEFF007080506D
00E095246737461725F373835364136
0FFF5246010000000000C003C20B5C2
2
```

Details:

LEN.	TYPE	VALUE
2	0x01	0x06
26	0xFF	0x4C0002150112233445566778 899AABBCCDDEEFF007080506D 0
14	0x09	0x5246737461725F3738353641 36
15	0xFF	0x5246010000000000C003C20B 5C22

HEX: 22C5

decimal: 8901
humidity: 89.01%

acceleration

HEX: 0BC2

decimal: 3010

temperature: 30.1°C

7 AT Commands

AT commands can be subdivided into two types:

Type	Command Format	Description
Query command	AT+[x]?	This command is used to query the current value of the returned parameter.
Setting command	AT+[x]=<...>	This command is used to set user-defined parameter value.

Note:

1. This command can be sent through APP.
2. Not every command has the above two types of commands.
3. The AT command must be **capitalized** and no need to be ended with a carriage return and line feed (CRLF).
4. If all parameters of the command are optional parameters, at least one parameter is needed to be filled, otherwise it is also regarded as a command error.

Example: AT+ADS=<0,1>,<0,1>,<10,10240>, you can fill in AT+ADS=,,500.

5. The parameter positions that are not filled in the optional parameter command must **be reserved**. Refer to the previous example.
6. Except for the user authentication function, all other setting commands are effective immediately and saved after power-off.

7. The master end must make MTU bigger than 128 bytes, otherwise, the commands cannot be set normally.

AT Command List

Table 3. AT Command List

Command	Function
AT+NAME	Query/set the device name.
AT+MAC	Query/set the MAC address.
AT+SERIAL_ID	Query/set the serial number.
AT+ADS	Query/set the broadcast parameters.
AT+POWER	Query/set the device transmit power.
AT+ENCRYPT	Query/set the encryption of the broadcast data.
AT+MODE	Query/set the working mode.
AT+CYCLE	Query/set the broadcast slots.
AT+DATA_CYCLE	Query/set the number of broadcast slots.

AT+CLR_CYCLE	Clear the broadcast slots.
AT+BEACON	Query/set the beacon-related parameters.
AT+ES	Query/set the Eddystone-related parameters.
AT+GSENSOR	Query/set the accelerometer sensor parameters.
AT+THINFO	Query/set the temperature and humidity sensor parameters.
AT+AUTH	Query/set the user authentication.
AT+VERSION	Query the device firmware version number.
AT+RESET	Restore the device parameters to factory settings and restart.
Command Return Value	
OK	Successful command
FAIL	Failed command
ERROR	Error command
BUSY	Busy command, please wait for the previous operation

Device Name

AT+NAME?	
Function	Query the device name.
Example	AT+NAME?
Return Value	AT+NAME=RFstar_XXXX OK
Remark	The command returns the device name correctly, the default is "RFstar_XXXX", XXXX is the last two bytes of the MAC address.

AT+NAME=	
Function	Set device name.
Example	AT+NAME=TEST-NAME
Return Value	OK
Remark	The maximum setting length cannot exceed 11 bytes.

MAC Address

AT+MAC?	
Function	Query the MAC address.
Example	AT+MAC?
Return Value	AT+MAC=8A:E5:84:7A:E7:C9 OK
Remark	The returned MAC address is in hexadecimal.

AT+MAC=	
Function	Set the MAC address.
Example	AT+MAC=F1:F2:F3:F4:F5:F6
Return Value	OK
Remark	The set new MAC address takes effective after reboot and can be saved after power off.

Serial Number

AT+SERIAL_ID?	
Function	Query the serial number.
Example	AT+SERIAL_ID?
Return Value	AT+SERIAL_ID=9853008 OK
Remark	The command returns the serial number correctly, the default is the last three bytes of the MAC address.

AT+SERIAL_ID=	
Function	Set the serial number.
Example	AT+SERIAL_ID=123456
Return Value	OK
Remark	The setting serial number is 123456. The setting range: 0 ~ 4294967295

Broadcast Parameter

AT+ADS?	
Function	Query the broadcast parameters.
Example	AT+ADS?
Return Value	AT+ADS=,1,200 OK
Remark	Parameter 1: Reserved Parameter 2: Device broadcast mode setting (0, un-connectable. 1, connectable). Parameter 3: Device broadcast interval in ms, the setting range: 100 ms ~ 10240 ms.

AT+ADS=	
Function	Set the broadcast parameters.
Example	AT+ADS=,0,500
Return Value	OK
Remark	Set to enable unconnectable broadcast with 500 ms interval, 1000 ms is by default. Note: Please set this parameter carefully, once it is set to un-connectable mode broadcast, any parameter is not allowed to be modified.

TX Power

AT+POWER?	
Function	Query the current transmit power.
Example	AT+POWER?
Return Value	AT+POWER=4 OK
Remark	The current transmit power of the device is 4 dBm.

AT+POWER=	
Function	Set the transmit power.
Example	AT+POWER=-10
Return Value	OK
Remark	The current transmit power of the device is -10 dBm. The device only supports the following 8 levels of transmit power: [-20, -10, -6, -4, -2, 0, 2, 4]. The default transmit power is 0 dBm .

Broadcast Data Encryption

AT+ENCRYPT?	
Function	Query the encryption of the broadcast data.
Example	AT+ENCRYPT?
Return Value	AT+ENCRYPT=0,11223344556677889911223344556677 OK
Remark	Parameter 1: 0: Disable broadcast data encryption. 1: Enable broadcast data encryption. Parameter 2: AES key (16 bytes).

AT+ENCRYPT=	
Function	Query broadcast data encryption.
Example	AT+ENCRYPT=1,11223344556677889911223344556677
Return Value	OK
Remark	Enable broadcast data encryption and set the key to "11223344556677889911223344556677", and the key must be 16 bytes. This encryption only encrypts data in iBeacon mode or UID mode. If enable encryption and then set TLM or URL, the Beacon will feedback FAIL. The range

of Beacon broadcast encryption includes UUID, Major, and Minor, with a total of 20 bytes, the first 16 bytes are encrypted by AES128, and the last 4 bytes are encrypted by XOR (the value of XOR is the last byte of MAC). The UID encrypted content includes 16 bytes of Namespace and Instance, which are encrypted by AES128.

Working Mode

AT+MODE?	
Function	Query the working mode.
Example	AT+MODE?
Return Value	AT+MODE=0 OK
Remark	0: iBeacon mode. 1: Eddystone mode.

AT+MODE=	
Function	Set the working mode.
Example	AT+MODE=1
Return Value	OK
Remark	Set the broadcast to Eddystone mode, iBeacon mode is by default.

Enable Broadcast Slots

AT+CYCLE?	
Function	Query the broadcast slots.
Example	AT+CYCLE?
Return Value	AT+CYCLE=0 OK
Remark	0: Disable the broadcast slots. 1: Enable the broadcast slots.

AT+CYCLE=1	
Function	Set the broadcast slots.
Example	AT+CYCLE=1

Return Value	OK
Remark	Enable the broadcast slots function. The default is disabled. After it is enabled, the broadcast slots can be preset through the commands " AT+BEACON " and " AT+ES " (up to 10 preset broadcast slots can be supported), and each broadcast slot will be set according to the preset broadcast cycle and transmit power.

Query Broadcast Slots Data

AT+DATA_CYCLE?	
Function	Query the broadcast slots data.
Example	AT+DATA_CYCLE?
Return Value	AT+DATA_CYCLE=2 OK
Remark	Query the slot number of the broadcast. 2 means there are two broadcast slots. The "AT+DATA_CYCLE=" can be used to get the broadcast data.

AT+DATA_CYCLE=1	
Function	Get the broadcast slot data.
Example	AT+DATA_CYCLE=1
Return Value	AT+DATA_CYCLE=1,020106.....,0,1000 OK
Remark	The first broadcast slot will be returned with the following format: Parameter 1: The serial number of the broadcast slot returned. 1 means the first broadcast slot. Use "AT+DATA_CYCLE?" to query the number of slots in total. (Range: 1 ~ 10) Parameter 2: Slot data. Parameter 3: The current TX power of this slot. 0 dBm for the example data. Parameter 4: The broadcast interval of this slot. 1000 ms for the example data.

Clear Broadcast Slots

AT+CLR_CYCLE	
Function	Clear the broadcast slots.
Example	AT+CLR_CYCLE
Return Value	OK

Remark	All broadcast slots are cleared.
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Beacon Parameters

AT+BEACON?	
Function	Query the Beacon parameters.
Example	AT+BEACON?
Return Value	AT+BEACON=004C,0807,0605,D0,0112233445566778899AABBCCDDEEFF0 OK
Remark	Parameter 1: Company ID, 2 bytes Parameter 2: Major, 2 bytes Parameter 3: Minor, 2 bytes Parameter 4: Reference RSSI at 1 m. This value is a signed type, and the negative values can be restored correctly by adding 1 after the reverse. Parameter 5: User-defined UUID data, 16 bytes.

AT+BEACON=	
Function	Set the Beacon parameters.
Example	AT+BEACON=4546,0102,0304,C2,0102030405060708090A0B0C0D0EF010,0,1000
Return Value	OK
Remark	Parameter 1: Set Company ID to 4546 Parameter 2: Set Major to 0102 Parameter 3: Set Minor to 0304 Parameter 4: Reference RSSI at 1 m is -62 dBm Parameter 5: User-defined UUID data is 0102030405060708090A0B0C0D0EF010 Parameter 6: Set the transmit power to 0 dBm. Parameter 7: Set the broadcast cycle to 1000 ms. Note: - The zero parameter cannot be omitted. For example, if you set Major to 1, you cannot fill in "1" or "01", but you should fill in "0001", otherwise ERROR will be returned. - Parameter 6 and parameter 7 only can be effective in broadcast slots mode (AT+CYCLE=1). They are used to set the TX power and broadcast interval for each slot. Those two parameters can be ignored if no need to be set respectively, and only

“AT+ADS” and “AT+POWER” can be used to set the related parameters, and “AT+BEACON” can not return those two parameters. If the broadcast slots mode is disabled, only “AT+ADS” and “AT+POWER” can be used to set the related parameters.

Eddystone Parameters

AT+ES?	
Function	Query the Eddystone parameters.
Example	AT+ES?
Return Value	UID Type: AT+ES=00,D0,11223344556677889900112233445566,0000 OK URL Type: AT+ES=10,C2,01,baidu,00 OK TLM Type: AT+ES=20,00,00,0D08,00,1AC0,0000003E,0000023A OK
Remark	Eddystone's three frame formats have different returns. For details, see the following analysis: UID Type: Parameter 1: UID frame fixed value, 0x00 Parameter 2: The reference RSSI at 1 m, which has been compensated by -41 dBm based on the RSSI at 0 m, range: -100 dBm ~ -7 dBm. This value is a signed type, and the negative values can be restored correctly by adding 1 after the reverse. Parameter 3: 16-byte UUID (Namespace + Instance) value Parameter 4: Reserved bytes, user can modify it. URL Type: Parameter 1: URL frame fixed value, 0x10 Parameter 2: The reference RSSI at 1 m, which has been compensated by -41 dBm based on the RSSI at 0 m, range: -100 dBm ~ -7 dBm. This value is a signed type, and the negative values can be restored correctly by adding 1 after the reverse. Parameter 3: URL prefix range 0x00 ~ 0x03 Parameter 4: URL content, users can directly access the URL by broadcasting the URL in a

	specific APP. Parameter 5: URL suffix, range 0x00 ~ 0x0D TLM Type: Parameter 1: TLM frame fixed value, 0x20 Parameter 2: Version number, must be 0x00 Parameter 3: When it is 0, it means that the voltage data is automatically obtained by the device; when it is 1, it means that the voltage is user-defined data Voltage data. Parameter 4: Voltage data. The default is automatically obtained by the device, in mv. Parameter 5: When it is 0, the temperature data is automatically obtained by the device, if it does not support the value, it is 0x8000; when it is 1, it means that the temperature is user-defined data. Parameter 6: Temperature data, in °C. Parameter 7: Broadcast count value, counting starts after power-on, reset can restore the count value. (This parameter only can be acquired, cannot be set.) Parameter 8: Time counting since power-on, resolving power is 100 ms, when the device is switched from the default iBeacon mode to Eddystone mode, the device will start counting from zero. Reset can restore the count value. (This parameter only can be acquired, cannot be set.)
--	---

AT+ES=	
Function	Set the Eddystone parameters.
Example	Different frame types of Eddystone have different formats, which are divided into the following three types: UID Type: AT+ES=00,D0,11223344556677889900112233445566,FF,0,1000 URL Type: AT+ES=10,D0,01,baidu,00,0,1000 TLM Type: AT+ES=20,00,00,0EC4,00,0A5A,0,1000
Return Value	OK
Remark	UID Type: Set the RSSI reference value of the UID frame to -48 dBm, the Namespace ID and Instance ID are set together, and set the reserved value to 0xFF. Set transmit power to 0 dBm and the broadcast cycle to 1000 ms.

URL Type:

Set the RSSI reference value of the URL frame to -48 dBm, the URL prefix is 0x01, baidu is the set URL content, and 00 is the URL suffix, which can be encoded as .com/. For other suffix format codes and prefix codes, please see the official below URL encoding table. Set transmit power to 0 dBm and the broadcast cycle to 1000 ms.

<https://github.com/google/eddystone/tree/master/eddystone-url>

TLM Type:

Parameter 4: "00" means that the voltage parameter of the TLM frame is automatically obtained by the device, then the parameter 0EC4 is considered invalid.

Parameter 6: "01" means setting the temperature data as user-defined data, then parameter 0A80 is considered valid. So 26.50 °C is the user-defined temperature. Set transmit power to 0 dBm and the broadcast cycle to 1000 ms.

Note:

1. You can switch between the three Eddystone modes only by setting the frame type, for example: Send the command "AT+ES=20,,,,," can directly switch to TLM mode, and send the command "AT+ES=00,,,,," can directly switch to UID mode.
2. The zero parameter cannot be omitted. For example, set the reserved byte of the UID frame to 000A, and you cannot fill in "0A" or "A", but should fill in "000A", and the device must be in Eddystone mode to query, otherwise it will return ERROR.
3. The transmit power and broadcast interval only can be effective in broadcast slots mode (AT+CYCLE=1). They are used to set the TX power and broadcast interval for each slot. Those two parameters can be ignored if no need to be set respectively, and only "AT+ADS" and "AT+POWER" can be used to set the related parameters, and "AT+ES" can not return those two parameters. If the broadcast slots mode is disabled, only "AT+ADS" and "AT+POWER" can be used to set the related parameters.

Accelerometer Sensor Parameters

AT+GSENSOR?	
Function	Query the accelerometer sensor parameters.
Example	AT+GSENSOR?
Return Value	AT+GSENSOR=2000,20,2 OK

Remark	Parameter 1: Accelerometer trigger threshold. Range: 1000 mg ~ 23000 mg
	Parameter 2: Accelerometer sensor sampling cycle. Range: 1 s ~ 65535 s
	Parameter 3: High-density broadcast duration. Range: 1 s ~ 65535 s
	The trigger threshold is set to 2000 mg, the sampling cycle to 20 s, and the high-density broadcast duration to 2 s.

AT+GSENSOR=	
Function	Set the accelerometer sensor parameters.
Example	AT+GSENSOR=2000,30,3
Return Value	OK
Remark	Set the trigger threshold to 2000 mg, sampling cycle to 30 s, and high-density broadcast duration to 3 s when the accelerometer is triggered.

Temperature and Humidity Sensor Parameters

AT+THINFO?	
Function	Query the temperature and humidity sensor parameters.
Example	AT+THINFO?
Return Value	AT+THINFO=10 OK
Remark	The temperature and humidity sensor sampling cycle is 10 s.

AT+THINFO=	
Function	Set the temperature and humidity sensor parameters.
Example	AT+THINFO=60
Return Value	OK
Remark	Set the temperature and humidity sensor sampling cycle to 60 s. Range: 1 s ~ 65535 s

User Authentication

AT+AUTH?	
Function	Query the current status of the user authentication function.
Example	AT+AUTH?
Return Value	AT+AUTH=0,0000,60

	OK
Remark	Parameter 1: Enable / disable the user authentication function. 0, disabled. 1, enabled. Parameter 2: Key, the key length cannot exceed 16 bytes. The default is "0000". Parameter 3: Authentication valid time (in s), 15 s is by default. Note: It will take effect after disconnection when enabled, and Bluetooth will be automatically disconnected if the authentication key is not received within the valid time. When the authentication takes effect, the password needs to be filled in for verification after each connection.

AT+AUTH=	
Function	Set the user authentication function.
Example	AT+AUTH=1,123456,20
Return Value	OK
Remark	Enable the user authentication function, set the key to "123456", and the effective time is 20 s. The setting range of the valid time is 1 s ~ 65535 s. The serial port setting takes effect immediately, and the APP setting takes effect after disconnection. Once the user authentication is enabled, a key is required for each connection, otherwise, the user authentication fails and all commands are invalid, FAIL information prompt will be returned. The recommended value should be 10 s or more, otherwise, it will cause the authentication time to be too short to cause authentication failure.

Firmware Version Number

AT+VERSION	
Function	Query the firmware version number.
Example	AT+VERSION?
Return Value	AT+VERSION= RF-IB-AT02-V0.1.6_20211020 OK

Restore Factory Settings

AT+RESET	
Function	Restore factory settings.

Example	AT+RESET
Return Value	OK
Remark	The device restarts immediately after the settings are successful.



8 Revision History

Date	Version No.	Description
2020.11.26	V0.1.0	The initial version is released.
2021.01.05	V0.1.1	Add the related AT commands of sensors and broadcast slots.
2021.03.22	V0.1.2	Add the button function specifications, and modify the broadcast slots function.
2021.08.03	V0.1.3	Add the encryption identifier in the response packet. Modify the function of clearing broadcast slots. Modify the setting error problem of RFU under broadcast slots mode. Modify the unchangeable RSSI returned when under TLM. Optimize the UUID related to Device information. Solve the power consumption error without sensors. Modify the broadcast name length from 14 bytes to 13 bytes. The default mode after power-on from shut down to boot up. After booting up, the device can be shut down by putting the button.
2021.08.30	V0.1.4	Update some parameters.
2021.09.06	V0.1.5	Update the dimension with the enclosure.
2021.10.27	V0.1.6	Add the battery capacity of broadcast data. Modify the broadcast name to 11 bytes.
2022.04.11	V0.1.6	Add the response packet data description. Add MAC address command. Add the Serial ID range.
2022.12.16	V0.1.6	Update the descriptions.

Note:

- The document will be optimized and updated from time to time. Before using this document, please make sure it is the latest version.
- To obtain the latest document, please download it from the official website: www.rfstariot.com and www.szrfstar.com.

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